

AI-Sentinel

Intelligent Technical Debt & Code Quality Analyzer

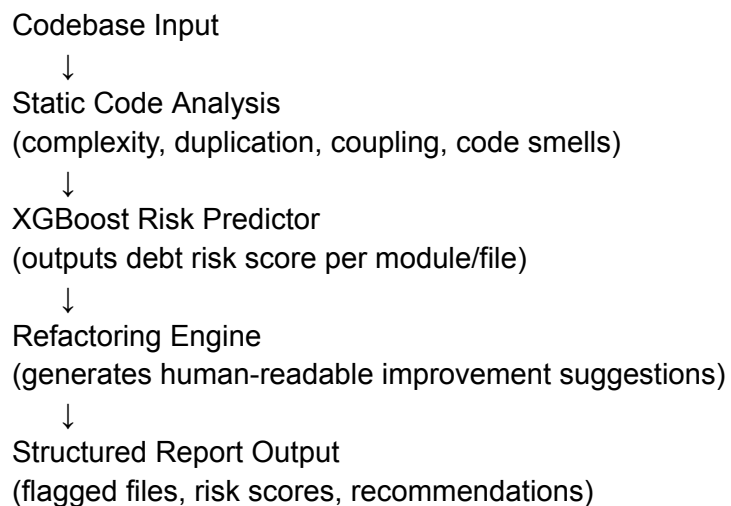
This project is actively under development. Core functionality is complete and has been evaluated in an academic setting. Further improvements to accuracy, UI, and pipeline robustness are in progress.

Overview

AI-Sentinel is an AI-powered system designed to identify and quantify technical debt in software codebases — one of the most costly and underdiagnosed problems in the software industry. Research estimates that developers lose approximately 23% of their productive time navigating poorly structured or suboptimal code.

AI-Sentinel addresses this by combining classical machine learning with large language model reasoning to deliver both risk scores and actionable refactoring suggestions — automatically.

Pipeline



Tech Stack:

- **Backend:** Python, FastAPI
 - **Static Analysis:** radon, Python ast
 - **ML Model:** XGBoost
 - **LLM:** Ollama + CodeLlama 13B (primary) / DeepSeek-Coder 6.7B (lightweight alternative)
 - **Frontend:** Gradio (demo), D3.js (heatmap)
 - **Database:** SQLite (dev) / PostgreSQL (production)
 - **CI/CD:** GitHub Actions
 - **Security Data:** NIST NVD REST API
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Features

- **Automated code scanning** — analyzes entire codebases for quality issues
 - **Risk scoring** — quantifies technical debt per file and module using ML
 - **LLM-powered suggestions** — generates specific, actionable refactoring guidance
 - **Structured reporting** — outputs findings in a clean, readable format for engineering teams
 - **Industry-oriented** — designed for real codebases, not toy examples
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Use Cases

- Engineering teams managing large or legacy codebases
 - Code review automation and pre-merge quality checks
 - Academic and research evaluation of software quality metrics
 - Developer productivity tooling
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